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Class : 12

Register Number | 1 | 2 | A | 3 | 1 |

COMMON QUARTERLY EXAMINATION - 2026-27

PHYSICS

Time Allowed : 3.00 Hours] [Max. Marks : 70

PART - I

I. **Choose the appropriate answer and write with option.** 15x1=15

1. The temperature coefficient of resistance of a wire is 0.00125 per °C. At 20°C, its resistance is 1Ω . The resistance of the wire will be 2Ω at
   a) 800 °C b) 700 °C c) 850 °C d) 820 °C
2. In an electromagnetic wave travelling in free space the rms value of the electric field is 3 V m^{-1}. The peak value of the magnetic field is
   a) 1.414×10^{-8} T b) 1.0×10^{-8} T c) 2.828×10^{-8} T d) 2.0×10^{-8} T
3. An inductor 20 mH, a capacitor 50 μF and a resistor 40Ω are connected in series across a source of emf n = 10 sin 340 t. The power loss in AC circuit is
   a) 0.76 W b) 0.89 W c) 0.46 W d) 0.67 W
4. An electric field $\vec{E} = 10x\,\hat{i}$ exists in a certain region of space, then the potential difference $V = V_0 - V_A$, where V_0 is the potential at the origin and V_A is the potential at x = 2 m is
   a) +10 V b) –20 V c) +20 V d) –10 V
5. The type of lens formed by a bubble inside water is
   a) convex b) concave c) plano - convex d) plano - concave
6. In a Wheatstone's bridge P = 100 Ω, Q = 1000 Ω and R = 50 Ω. If the galvanometer shows zero deflection, the value of S is,
   a) 0.5 Ω b) 50 Ω c) 500 Ω d) 5000 Ω
7. A circular coil of radius 5 cm and 50 turns carries a current of 3 ampere. The magnetic dipole moment of the coil is nearly.
   a) 1.0 Am^2 b) 1.2 Am^2 c) 0.5 Am^2 d) 0.8 Am^2
8. For light incident from air on a slab of refractive index 2, the maximum possible angle of refraction is
   a) 30° b) 45° c) 60° d) 90°
9. If the amplitude of electric field for an electromagnetic wave is 300 Vm^{-1}, then amplitude of magnetic field is
   a) 10^{-6} T b) 3×10^{-6} T c) 10^{-6} T d) 3×10^{-6} T
10. The speed of light in an isotropic medium depends on
    a) its intensity b) its wavelength
    c) the nature of propagation d) the motion of the source with respect to medium
11. In a series RL circuit, the resistance and inductive reactance are the same. Then the phase difference between the voltage and current in the circuit is
    a) $\frac{\pi}{4}$ b) $\frac{\pi}{2}$ c) $\frac{\pi}{6}$ d) zero
12. The potential energy of magnetic dipole whose dipole moment is $\vec{P}_m = (-0.5\,\hat{i} + 0.4\,\hat{j})\ Am^2$ kept in uniform magnetic field $\vec{B} = 0.2\,\hat{i}$ T.
    a) -0.1 J b) -0.8 J c) 0.1 J d) 0.8 J
13. A parallel plate capacitor stores a charge Q at a voltage V. Suppose the area of the parallel plate capacitor and the distance between the plate are each doubled, then which is the quantity that will change?
    a) capacitance b) charge c) voltage d) energy density
14. The potential due to an electric dipole varies as
    a) $\frac{1}{r}$ b) $\frac{1}{r^2}$ c) $\frac{1}{r^3}$ d) $\frac{1}{r^4}$
15. Unit of self inductance is
    a) Wb A^{-1} b) VAs^{-1} c) Hs^{-1} d) all the above

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PART - II

II. Answer Any six questions. Question Number 24 is Compulsory. **6x2=12**

16. Is an ammeter connected in series or parallel in a circuit? Why?
17. Give any two uses of microwaves.
18. Find the dispersive power of a prism if refractive indices of flint glass for red, green and violet colours are 1.613, 1.620 and 1.632 respectively.
19. What are the properties of an equipotential surface?
20. State Fleming's right hand rule.
21. Define electrical resistivity.
22. Why does sky appear blue?
23. What is magnetic susceptibility?
24. A square coil of side 30 cm with 500 turns is kept in a uniform magnetic field of 0.4T. The plane of the coil is inclined at an angle of 30^0 to the field. Calculate the magnetic flux through the coil.

PART - III

III. Answer Any six questions. Question Number 33 is Compulsory **6x3=18**

25. Mention the various energy losses in a transformer.
26. Derive an expression for electrostatic potential due to a point charge.
27. Calculate the resultant capacitance for the following combination of capacitors.



28. Obtain the equation for apparent depth.
29. State Kirchhoff's current and voltage rule.
30. Explain the types of absorption spectrum.
31. Discuss about series connection of cells.
32. Give an account of magnetic Lorentz force.
33. What is the focal length of the combination if the lenses of focal length -70 cm and 150 cm are in contact? What is the power of the combination?

PART - IV

IV. Answer all Questions. **5x5=25**

34. a) Obtain the expression for electric field due to an infinitely long charged wire.

(OR)

b) Explain in detail the effect of a dielectric placed in a parallel plate capacitor when the capacitor is disconnected from the battery.

35. a) Describe the microscopic model of current and obtain microscopic form of Ohm's law.

(OR)

b) Obtain the condition for bridge balance in Wheatstone's bridge.

36. a) Explain the working of a single-phase AC generator with necessary diagram.

(OR)

b) Find out the phase relationship between voltage and current in an pure inductive circuit.

37. a) Derive the equation for refraction at single spherical surface.

(OR)

b) Obtain the equation for radius of illumination (or) Snell's window.

38. a) (i) State Ampere's circuital law.
(ii) Compute the magnitude of the magnetic field of a long straight wire carrying a current of 1A at distance of 1 m from it. Compare it with Earth's magnetic field.

(OR)

b) (i) Why are electromagnetic waves non-mechanical?
(ii) If the relative permeability and relative permittivity of a medium are 1.0 and 2.25 respectively, find the speed of the electromagnetic wave in this medium.

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